

CodSoft AI Internship - Task 2 Report

Task Name: Unbeatable Tic-Tac-Toe AI | Domain: Artificial Intelligence

1. Objective & Problem Overview

Develop an unbeatable Artificial Intelligence agent to play Tic-Tac-Toe against a human user. The program evaluates all theoretical game board outcomes using decision tree search algorithms.

2. System Architecture & Methodology

The game engine employs the Minimax recursive backtracking algorithm. The algorithm treats the AI as a maximizing player (+1 for win) and the human user as a minimizing player (-1 for loss). By recursively searching every move combination down to terminal states (win/loss/draw), the AI guarantees an optimal decision, making it unbeatable.

3. Python Source Code Implementation

```
import math

def check_winner(board):
    for r in board:
        if r[0] == r[1] == r[2] and r[0] != " ": return r[0]
    for c in range(3):
        if board[0][c] == board[1][c] == board[2][c] and board[0][c] != " ": return board[0][c]
    if board[0][0] == board[1][1] == board[2][2] and board[0][0] != " ": return board[0][0]
    if board[0][2] == board[1][1] == board[2][0] and board[0][2] != " ": return board[0][2]
    return None

def is_full(board):
    return all(cell != " " for row in board for cell in row)

def minimax(board, depth, is_max):
    winner = check_winner(board)
    if winner == "O": return 1
    if winner == "X": return -1
    if is_full(board): return 0

    if is_max:
        best = -math.inf
        for i in range(3):
            for j in range(3):
                if board[i][j] == " ":
                    board[i][j] = "O"
                    best = max(best, minimax(board, depth + 1, False))
                    board[i][j] = " "
            return best
    else:
        best = math.inf
        for i in range(3):
            for j in range(3):
                if board[i][j] == " ":
                    board[i][j] = "X"
                    best = min(best, minimax(board, depth + 1, True))
                    board[i][j] = " "
            return best

    return best
```

4. Sample Execution & Test Output

```
Tic-Tac-Toe AI - You are X, AI is O
| |
-----
Enter row (0-2): 1
Enter col (0-2): 1

AI is making a move...
O | |
-----
| X |
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Outcome: AI evaluates all possibilities, securing a win or forced tie.
```